Layer 7: Enterprise Governance & Multi-Tenancy Plane [Control Catalog & Operator API]

- **FastAPI Operator API** — REST Endpoints & Token Auth
- **Role-Based Access Control** — RBAC Engine (Viewer/Approver/Admin)
- **Tenant Registry & Isolation** — Multi-Tenant Quota Manager
- **Streaming Keyset Audit** — O(1) Memory Compliance Exporter

Governance State & Keyset Audit Stream

Layer 6: Orchestration, Concurrency & Distributed Control [Distributed Consensus & Locks]

- **Asynchronous ControlLoop** — Periodic Scanner & Resumability
- **Two-Tier Conflict Arbitration** — Priority: Recovery > Schema > Opt
- **PostgreSQL Advisory Locks** — Target-Level Non-Blocking Locks
- **Durable Global Kill Switch** — Sub-Tick State Freeze & Heartbeat

Operator Escalation
Human Approval Queue
(Sign-off & Kill Switch)

Two-Tier Lock Arbitration & Concurrency Verification

Layer 5: Safe Execution & Graduated Autonomy Engine [Write-Ahead Audit & Dispatch]

- **Write-Ahead Audit Log** — Pre-Execution State (WAL Engine)
- **Tri-Mode Autonomy Ladder** — Shadow | Approval | Autonomous
- **Action Dispatcher & Handlers** — Airflow, Kafka & Database Mutators
- **Bounded Retry & Escalation** — 3-Attempt Limit & Operator Fallback

PolicyDecision (d) [OPA Invariant Verdict]

Layer 4: Declarative Policy Gate & Containment Plane [Open Policy Agent (OPA)]

- **Open Policy Agent** — Deterministic OPA Rego Engine
- **Rate-Limit Runaway Guard** — Max 5 Operations / 10 Min Window
- **Marginal Cost Budget Guard** — Marginal Delta C <= Budget Bound
- **Fail-Safe Escalation Default** — Deny-by-Default Fallback Guarantee

ProposedAction (a) [Closed-Vocabulary Contract]

Layer 3: Bounded Multi-Agent Supervisory Tier [Closed-Vocabulary Reasoning]

Cognitive Invariants: Closed Pydantic Action Schema | Deterministic Temp = 0 | Per-Incident Token Limits

- **Monitoring Agent** — Fast-Tier LLM + Z-Score Pre-Filter
- **Optimization Agent** — Reasoning-Tier Pool Tuning
- **Schema Agent** — Breaking Drift & Quarantine
- **Recovery Agent** — Task Replay & Partition Rollback

TelemetrySnapshot (S_t)

Layer 2: Telemetry & Observability Subsystem [Pipeline State & Metrics]

- **Telemetry Collector Engine** — Airflow REST Probes & OS Metrics
- **Data Freshness SLA Engine** — Stream Latency & Batch Staleness
- **Normalized Hybrid Cost Ledger** — Step-Integrated Compute & Storage
- **Failure & Incident Tracker** — Incident State & MTTR Ledger

Raw Metrics & Observation Stream

Layer 1: Pipeline Data Plane & Connector Boundary [Execution Substrate & Storage]

- **Connector Protocol Boundary** — Airflow / Prefect / Noop Plugs
- **Batch Workload DAGs** — ETL Pipelines & TPC-DS Benchmarks
- **Streaming Event Bus** — Redpanda / Kafka Window Aggregations
- **Transactional Partition Manager** — Zero-Copy O(1) Rollback Pointer Flip

Gated Actuation Loop
Safe Data Plane Side-Effects
(Replay / Rollback / Sizing)